

MODULE-5:

Here are the main topics and important points from the AWS training document:

Module 5: Networking and Content Delivery

This module provides a detailed guide to networking and content delivery within the AWS cloud environment. It focuses on key AWS services like Amazon VPC, Amazon Route 53, and Amazon CloudFront.

Section 1: Networking Basics

This section reviews fundamental networking concepts:

- **Computer Networks:** Networks consist of two or more connected client machines that share resources, often logically partitioned into subnets and facilitated by networking devices like routers and switches.
- **IP Addresses:** Each networked machine has a unique IP address for identification. IPv4 addresses are 32-bit numerical labels in decimal format (e.g., 192.0.2.0). IPv6 addresses are 128-bit and accommodate more devices.
- **CIDR Notation:** Classless Inter-Domain Routing (CIDR) is used to describe networks and IP address ranges. It indicates the network address and the number of fixed bits in the routing prefix (e.g., 192.0.2.0/24).
- **OSI Model:** The Open Systems Interconnection (OSI) model is a conceptual framework with seven layers that explain how data travels across a network. It shows protocols and addresses used at each layer, helping understand communication in networks and VPCs.

Section 2: Amazon VPC

This section introduces Amazon VPC, a service for provisioning logically isolated virtual networks within the AWS cloud:

- **Amazon VPC Features:** Amazon VPC enables users to define their own virtual networks within AWS, controlling IP address ranges, subnets, route tables, network gateways, and security layers.
- **VPCs and Subnets:** VPCs are isolated and dedicated to specific AWS accounts within a single AWS Region but can span multiple Availability Zones. They are divided into subnets, each with a range of IP addresses and residing in a single Availability Zone. Subnets are classified as public (internet access) or private (no direct internet access).
- **IP Addressing:** VPCs are assigned an IPv4 CIDR block that cannot be changed later. Subnet CIDR blocks must not overlap and duplicate IP addresses within a VPC. IPv6 is also supported.
- **Reserved IP Addresses:** AWS reserves five IP addresses per subnet for specific functions like network address, router communication, DNS, future use, and broadcast.
- **Public IP Addresses:** Instances in a VPC get a private IP address automatically. They can obtain a public IP address manually (Elastic IP address) or automatically through subnet settings.

- **Elastic IP Addresses:** Elastic IP addresses are static public IPv4 addresses associated with an AWS account. They allow remapping to different instances for masking failures.
- **Elastic Network Interfaces:** These virtual interfaces can be attached to or detached from instances within a VPC. They retain their attributes when moved between instances.
- **Route Tables and Routes:** Route tables define rules for directing network traffic from subnets. Each route specifies a destination CIDR block and a target. Each subnet is associated with one route table.

Section 3: VPC Networking

This section details various networking options within a VPC:

- **Internet Gateways:** These components allow communication between instances in a VPC and the internet, acting as a target in route tables for internet traffic and performing network address translation.
- **NAT Gateways:** NAT gateways enable instances in private subnets to connect to the internet or AWS services without allowing incoming connections from the internet. They reside in public subnets and are associated with an Elastic IP address. NAT instances can be used but offer less availability and require more administration.
- **VPC Sharing:** This feature allows sharing subnets with other AWS accounts within the same AWS Organization, enabling multiple accounts to utilize a centrally managed VPC.
- **VPC Peering:** VPC peering establishes private networking connections between VPCs within the same AWS account, across accounts, or across Regions. It requires non-overlapping IP address ranges and does not support transitive peering (direct connection needed for each VPC pair).
- **AWS Site-to-Site VPN:** This service creates VPN connections between a VPC and a remote network by utilizing a virtual private network (VPN) gateway attached to the VPC and a customer gateway (representing the VPN device).
- **AWS Direct Connect:** AWS Direct Connect establishes a dedicated private network connection between a network and a Direct Connect location, offering reduced costs, increased bandwidth, and consistent performance.
- **VPC Endpoints:** VPC endpoints enable private connections between a VPC and supported AWS services or VPC endpoint services using AWS PrivateLink. They eliminate the need for internet gateways, NAT devices, or VPN connections.
- **AWS Transit Gateway:** This service simplifies network management by acting as a central hub for connecting multiple VPCs, data centers, or remote offices. It provides a hub-and-spoke model, reducing complexity and operational costs compared to VPC peering.

Section 4: VPC Security

This section focuses on security aspects within a VPC:

- **Security Groups:** Security groups act as virtual firewalls at the instance level, controlling inbound and outbound traffic. They contain rules that filter traffic and are stateful, meaning responses to allowed traffic are automatically permitted.
- **Network Access Control Lists (Network ACLs):** Network ACLs provide an optional security layer at the subnet level, controlling traffic in and out of subnets. They contain numbered rules that

allow or deny traffic and are stateless, requiring explicit rules for both incoming and outgoing traffic.

- **Security Groups vs. Network ACLs:** Security groups operate at the instance level and are stateful, while network ACLs operate at the subnet level and are stateless. Security groups only allow rules, while network ACLs allow both allow and deny rules.

Section 5: Amazon Route 53

This section covers Amazon Route 53, a highly available and scalable cloud DNS service:

- **Amazon Route 53 Features:** Route 53 translates domain names into IP addresses, routing users to internet applications. It supports IPv4 and IPv6, connects requests to AWS and external infrastructure, performs health checks, manages traffic flow, and allows domain name registration.
- **DNS Resolution:** Route 53 resolves domain names by checking its records and returning the corresponding IP address to the DNS resolver.
- **Supported Routing Policies:** Route 53 supports various routing policies, including simple, weighted round robin, latency, geolocation, geoproximity, failover, and multivalue answer routing.
- **Multi-Region Deployment:** Route 53 facilitates multi-region deployments, directing users to the nearest Elastic Load Balancing load balancer for improved performance.
- **DNS Failover:** Route 53 improves application availability through configuring backup scenarios, enabling multi-region architectures, and using health checks to monitor resources and redirect traffic during outages.

Section 6: Amazon CloudFront

This section introduces Amazon CloudFront, a content delivery network (CDN) service:

- **Content Delivery and Latency:** Network performance is crucial for content delivery, as requests travel through multiple networks and hops, impacting latency and responsiveness. CDNs address these challenges.
- **Content Delivery Network (CDN):** A CDN uses globally distributed caching servers to store copies of frequently requested content, delivering content from the nearest server to the user, reducing latency, and improving performance.
- **Amazon CloudFront Features:** CloudFront is a fast, global CDN that securely delivers data, videos, applications, and APIs. It utilizes edge locations and regional edge caches for optimized content delivery. It offers a self-service model with pay-as-you-go pricing.
- **CloudFront Infrastructure:** CloudFront uses edge locations (data centers serving popular content) and regional edge caches (storing less frequently accessed content) to ensure low latency and efficient content delivery.
- **CloudFront Benefits:** CloudFront is fast, global, secure, programmable, deeply integrated with AWS, and cost-effective.
- **CloudFront Pricing:** CloudFront charges are based on data transfer out, HTTP(S) requests, invalidation requests, and dedicated IP custom SSL certificates.

The document includes hands-on activities, labs, and a knowledge check to reinforce the concepts covered.